

LAB ONE:

Sin-wave plotted in time domain would result in a strength of signal (volts) over time. In MATLAB we can use the oscilloscope in Simulink to do this for us. In frequency domain, we would plot the strength over frequency which a spectrum analyser would help in matlab.

Spectrum analyser does this using a maths technique called a Fast Fourier Transform (FFT).

Now for the Lab itself, open Simulink:

Digital clock → found in library browser → Simulink (left menu) → sources. This block helps us to decide the clock rate → which we can control the sampling rate of the simulation. (20kHz is standard (1/20000))

Signal Generator → again in sources. Amplitude to 1 and frequency to 100 (units to Hertz).

Oscilloscope → Library browser > DSP System Toolbox > Sinks > Time scope.

Now connect digital clock to signal generator to time scope. To use its tools, 'cursors measurements' (ruler icon).

By using the time scope, we can get the signals amplitude, period and frequency.

Now to analyse of frequency domain of the signal. Done by using a spectrum analyzer. When you change the frequency of the sine wave in the Signal Generator, the peak in the frequency domain (on the Spectrum Analyzer) moves left or right, matching the new frequency. If you set it to 3000 Hz, you'll see the peak at 3 kHz, for example Changing the amplitude affects the height of the peak—a higher amplitude results in a taller peak because the signal has more power. The maximum value of the horizontal axis is 10 kHz because the sampling rate is 20 kHz, and according to the Nyquist theorem, the highest frequency that can be accurately displayed without aliasing is half the sampling rate, which is 10 kHz.

LAB TWO:

When you switch the Signal Generator to a square wave (1 kHz frequency, 1V amplitude), the Frequency Domain shows something that's not obvious in the Time Domain:

You don't just see a single frequency like with a sine wave—you see multiple peaks at odd harmonics of the base frequency (1 kHz).

Using the data cursors, you should notice peaks at:

- 1 kHz (fundamental frequency)
- 3 kHz
- 5 kHz
- 7 kHz
- 9 kHz

These are the odd-numbered harmonics, and they decrease in strength as frequency increases. This shows that a square wave is made up of a sum of sine waves at these frequencies, which isn't obvious from just looking at the square wave in the Time Domain.

Remember that a square wave is actually composed of a number of different signals which are all sine-waves at various frequencies which each frequency present is related to fundamental frequency (1kHz). These frequencies relate to each other by integer multiples are called harmonics, or harmonically related frequencies.

Even harmonics -> Even integer multiples.

Odd harmonics -> Odd integer multiples.

For a square waveform:

The additional frequencies (harmonics) are odd multiples of the fundamental frequency. So if the fundamental is 1 kHz, the harmonics appear at 3 kHz, 5 kHz, 7 kHz, etc. These are called odd harmonics.

The power values (height of the peaks) decrease as the frequency increases. Each harmonic has less power than the one before it. This means the higher harmonics are weaker, which is why the square wave still has a shape, but not perfectly sharp edges.

4.1 Relative amplitudes of harmonics:

The theory of the square wave predicts the relative amplitudes of the harmonics. Like for example compared to the fundamental (first harmonic), the third harmonic is at 1/3 of the voltage.

so the voltage ratio can be found using $10^{\frac{r-s}{20}}$ in which r is the power of the fundamental frequency and the power of the nth harmonic is s (r and s are in dbm units). (for example r is 29

and s is 19 for the first and third harmonic, and using the voltage ratio it would be 3.16 which makes sense without theory). Like the third harmonics voltage would be $1/3.16$ so yes! Makes sense!

Measurements:

Fundamental peak: 29.06

3rd harmonic: 19.44

5th harmonic: 15.11

7th harmonic: 11.90

So here are the ratios:

- 3.16
- 5.01
- 7.07

LAB THREE:

Signal generators produce nice and tidy signals with clear max and min voltages which is not very real. In real life RF receivers receive attenuated version of the transmitted signal and the received signal suffers from locally generated noise, hence getting corrupted.

Random noise source > can be found in DSP system toolbox > sources > random source. Type should be gaussian.

A sum block (adds two signals) > Simulink > commonly used block > sum. (Sum block has two inputs and one output)

After doing the lab, here are the analysis of only the noise first:

- Time Domain (Oscilloscope):
The waveform looks completely random, with fast, unpredictable changes—no repeating pattern like a sine or square wave.
- Frequency Domain (Spectrum Analyzer):
Unlike sine waves (which show sharp peaks at specific frequencies), white noise shows a flat, wide spectrum. This means it has power spread evenly across all frequencies (up to the Nyquist limit).

Why is it called white noise?

Just like white light contains all visible light frequencies mixed together, white noise contains all audible frequencies with equal power. That's why the spectrum looks flat. It includes everything, not just one tone or a few harmonics.

Now let's do white noise and the signal as well.

- Time Domain (Oscilloscope):
now we can see the sine wave but with much more noise.
- Frequency Domain (Spectrum Analyzer):
A clear peak at 1 kHz appears — this is your sine wave. The flat noise floor is still present across all frequencies (like before), but now the sine wave stands out above it.

So the difference between these two is that the only noise, there was no peak (at 1kHz which is the sine wave) and just flat spectrum.

About Grass, which is the wavy bottom line in the spectrum analyzer is called grass which is basically noise. By changing the variance, the higher it goes the bigger and taller and more erratic the noise and the grass gets and with a lower variance the grass is flatter and smoother. So, we can say that the variance is sth that controls the noise power.

LAB FOUR:

A basic fact in communications is that simple periodic signals cannot convey information. This means that if we don't change anything (period, amplitude, frequency...) no information is conveyed. So, we have to modulate one (or more) of the basic characteristics of the signal. The RF signal is often referred to as the carrier signal, and the message or information signal is known as the Baseband Signal. (Baseband is used since frequency is generally much lower than the carrier frequency).

Some simple modulations would be:

- Amplitude modulation, AM (changing magnitude of a signal)
- Frequency modulation, FM (changing frequency of a signal)
- Phase modulation, PM (changing phase of a signal)

The carrier signal amplitude varies between two values. The frequency of the amplitude variation can be seen as sth like the envelope of the signal (information signal) amount of variation depends on relative size of the fixed component of the carrier signal and the varying component.

A_c is amplitude of fixed component A_m is amplitude of varying component.
 A_c / A_m -> known as modulation index, m .

m can be measured directly in time domain. E_{max} is maximum (ptp) amplitude of carrier signal. E_{min} is minimum (ptp) amplitude. We can measure m again as:

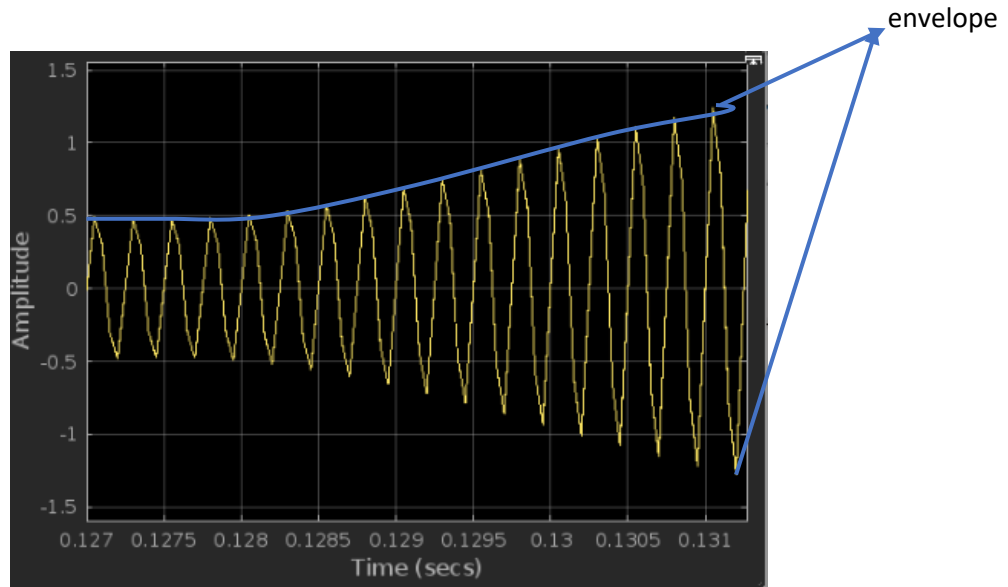
$$m = \frac{E_{max} - E_{min}}{E_{max} + E_{min}}$$

AM signal modulated with $m < 1$ -> we can recover the information signal by detecting the carrier envelope. Now we only need to find either the + or – envelope edges. (Both represent the original information signal)

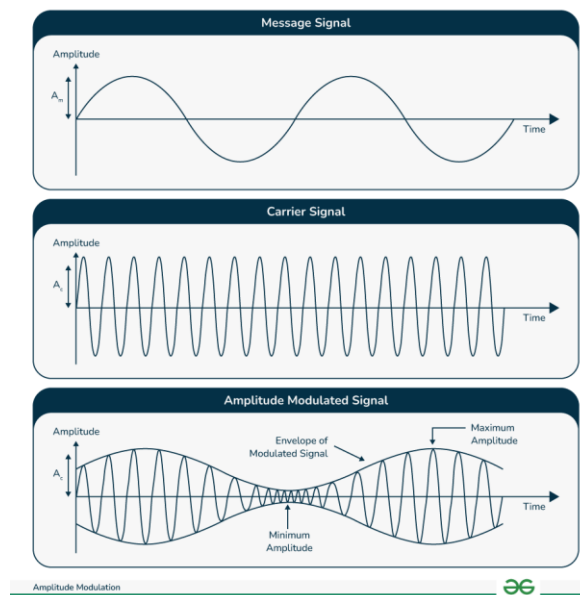
If the ratio, m , becomes bigger than unity $>$ signal is over-modulated. This means that the signals cannot be demodulated properly by simply detecting the envelope. (also causes some other issues like producing frequencies outside allocated radio channel)

Know that in real life the carrier signal has a very high frequency (GHz or THz).

Lab results: information signal (0.5 amplitude and 100 frequency) and carrier signal (1 amplitude and 4kHz frequency)



Amplitude Modulation



Carrier	A fast wave that helps us "send" data
Message	The actual data (like a voice signal)
Modulation	Changing the carrier to match the message

Envelope	The shape traced by the peaks of the modulated signal
Why it matters	It visually shows your data in the time-domain plot

The frequency of the envelope would be 100Hz (or same as the information signal). Also by changing the amplitude we are changing the envelope and how fast they change. (no amplitude no envelope) now trying to find out E_{max} , E_{min} . $E_{max}=2.86$ and $E_{min}=0.95$ and so $m = 0.5$ and using A_m and A_c where the first one is 0.5 and the second one is 1 we still get the same m !

LAB FIVE:

When you modulate a carrier with a sine-wave information signal, you create two new frequencies in addition to the carrier.

These new frequencies are called sidebands:

- One at carrier + message frequency (upper sideband)
- One at carrier – message frequency (lower sideband)

So, if the carrier is 4 kHz and the message is 100 Hz, the AM signal contains:

- 4 kHz (carrier)
- 4.1 kHz (upper sideband)
- 3.9 kHz (lower sideband)

Lab result:

$$f_{USB} = f_c + f_m$$

$$f_{LSB} = f_c - f_m$$

f_m is the frequency of the information signal and f_c is the frequency of the carrier signal.

Example 1: $f_c=4\text{kHz}$ and $f_m=1\text{kHz}$

therefore upper sideband = 5kHz and lower sideband = 3kHz.

Example 2: $f_c=4\text{kHz}$ and $f_m=2.5\text{kHz}$

therefore upper sideband = 6.5kHz and lower sideband = 1.5kHz.

Sideband amplitudes grow with modulation index.

- At 0% ($m = 0$) $\rightarrow$ only the carrier appears.
- At 100% ($m = 1$) $\rightarrow$ strong sidebands at $f_c \pm f_m$ still symmetric.

Sideband frequencies stay the same (only amplitudes change).

The bandwidth (BW) of a standard AM signal is: $BW = 2 \times f_m$

- It depends only on the message (info) frequency, *not* the carrier.
- For info = 1 kHz $\rightarrow$ BW = 2 kHz
- For info = 2.5 kHz $\rightarrow$ BW = 5 kHz

The carrier frequency just shifts the whole thing in the spectrum; it doesn't affect the bandwidth. Basically a bandwidth would be the difference between upper sideband and lower

LAB SIX:

Frequency spectrum of a square wave contains many harmonics of the fundamental frequency. This will later help when we modulate a carrier with a signal (that is digital) (a train of pulses representing 1s and 0s)

When we use square waves to AM a signal, it is called Amplitude Shift Keying (ASK). If the depth is 100%, that means that the carrier amplitude is 0 during low and 1 during high ----> commonly known as On-Off Keying (OOK)

Modulation depth is another name for modulation index (the ratio in lab four)

Lab result:

Time domain analysis: a high-frequency carrier wave (amplitude changing from highs to lows) modulation controlled by square wave at 100 Hz (sharp transitions between max and min A) -----> characteristics of ASK and OOK cause the carrier completely drops out during low part.

Frequency domain analysis: a wide spread of spectral components, with symmetrical sidebands around carrier frequency (4kHz). The sidebands are multiple harmonics. -----> it is expected because a square wave is made up one

fundamental frequency (100Hz) and odd harmonics (300Hz, 500Hz, etc.). each of these harmonics create their own pair sidebands in the AM spectrum (hence why the plot is dense and wide)

A square wave contains frequencies and in modulation, each frequency component of the square wave generates a pair of sidebands (one above, one below) which is why the bandwidth is widen drastically.

Implications for Bandwidth:

So, the bandwidth is much larger when we modulate square waves. Bad for communication efficiency (BW is limited and regulated). Transmitting square-wave directly causes spectral spreading (making it hard to avoid interference with other channels)

Solutions:

1. Filtering
2. Digital modulation schemes (bandwidth efficient):
PSK, QAM, FSK
3. Compression or encoding techniques (reduce data rate before modulation)